DOCUMENTATION ON DAY 1 TASK

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1. INSTALLED TOOLS

The following development tools were verified and available on my system:

• Python

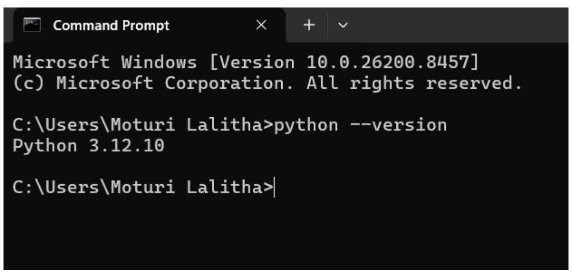

Figure 1: Python Installation Verification

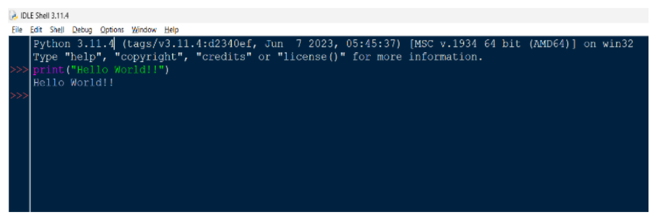

Figure 2: Python IDLE Execution

• Visual Studio Code (VS Code)

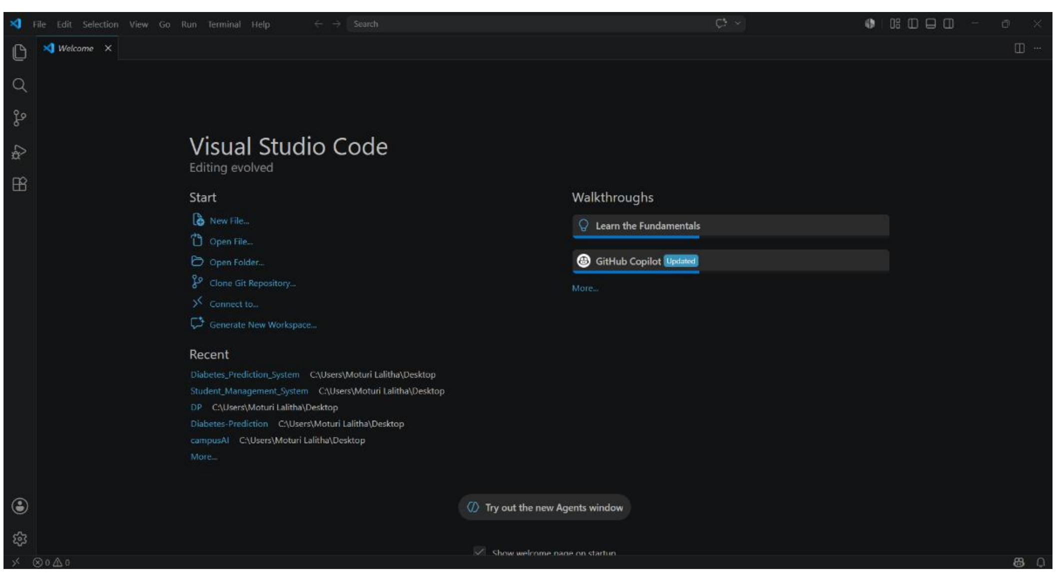

Figure 3: Visual Studio Code

• **Git**

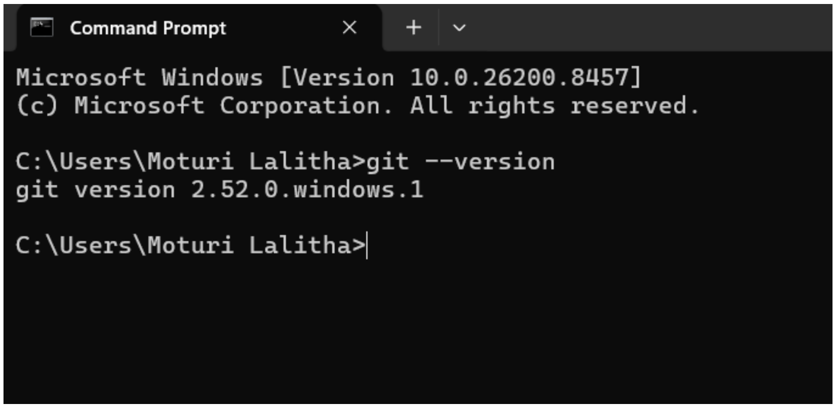

Figure 4: Git Installation Verification

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2. CREATE & SETUP GITHUB ACCOUNT

GitHub account was successfully created/verified. GitHub will be used throughout the cohort program for project management, version control, collaboration, and documentation.

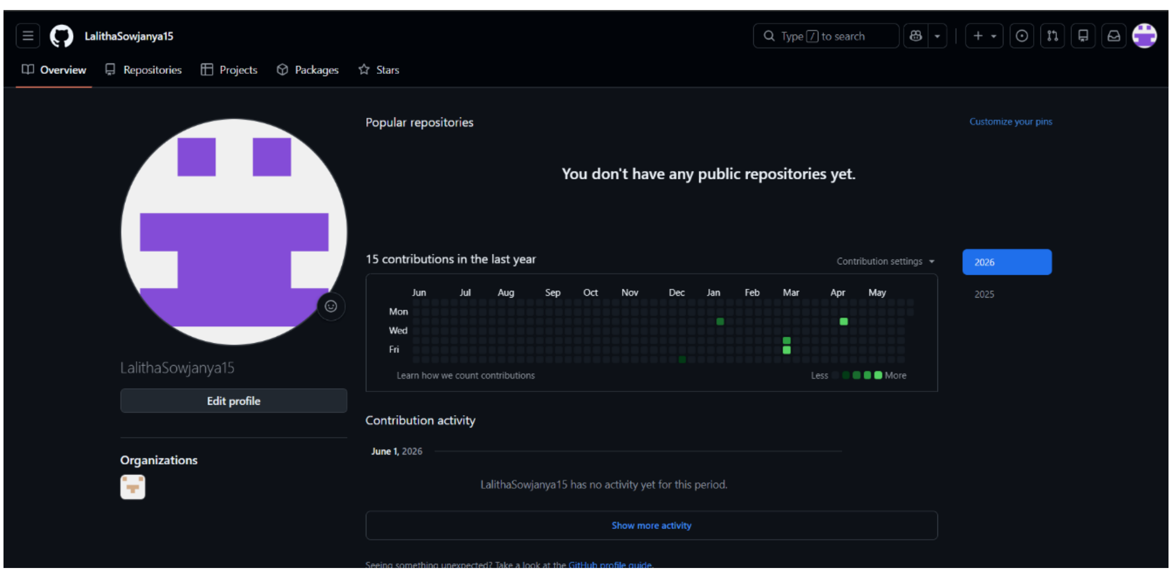

Figure 5: GitHub Profile

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3. RESEARCH TOPICS

3.1 Small Language Model (SLM)

Definition:

A Small Language Model (SLM) is a lightweight Artificial Intelligence language model designed to understand and generate human language while using fewer computational resources than Large Language Models (LLMs).

Features of SLM:

- Smaller model size
- Faster execution
- Lower memory usage
- Cost-effective deployment
- Can run on personal computers and laptops
- Suitable for educational and prototype projects

Limitations:

- Less knowledge compared to larger models
- Lower reasoning ability for complex tasks
- May provide less detailed responses

Examples:

• Phi • Tiny Llama • Small versions of Gemma • Small versions of Llama models

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3.2 Retrieval-Augmented Generation (RAG)

Definition:

Retrieval-Augmented Generation (RAG) is an AI architecture that retrieves relevant information from external sources such as PDFs, documents, databases, and websites before generating a response through an LLM or SLM.

Features of RAG:

- Uses external information sources
- Improves response accuracy
- Reduces hallucinations
- Supports document-based question answering
- Provides access to updated information

How RAG Works:

User (Asks a question)

↓

Retrieve Relevant Information (External source)

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Provide Context to LLM/SLM

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Generate Response

Advantages:

- More accurate responses
- Access to private documents
- No need for retraining when new documents are added

Limitations:

- Depends on retrieval quality
- Response depends on information quality

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4. PROBLEMS FACED DURING SETUP

No major issues were encountered during the setup process. Python, VS Code, Git, and GitHub were already available and functioning properly on the system.

5. WHAT I LEARNED TODAY

- Learned the purpose and importance of Python, VS Code, and Git in software development.
- Understood the role of GitHub in project collaboration.
- Learned the concept of Small Language Models (SLMs).
- Understood the differences between LLMs and SLMs.
- Learned the concept of Retrieval-Augmented Generation (RAG).
- Understood how RAG improves AI responses using external information sources.
- Learned the basic workflow of integrating AI models into applications.

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6. CONCLUSION

This task provided an introduction to the development tools and AI concepts that will be used during the cohort program. Through this activity, I gained a foundational understanding of SLMs, RAG systems, GitHub workflow, and development setup, which will help in building future AI-based projects.